

Mathematical Derivations of Diagnostic Thresholds

Five-Step IC Validation Framework — Revision 3

Key Revision (v3): Corrected Bonferroni multiple-testing threshold (Step 3 requires 11/11 positive windows, not 10/11); renamed Kelly–Malamud P/T to “IC Signal Fraction”; added conceptual-shift clarification for the daily positive fraction test; marked preliminary thresholds as empirically calibrated. See §9 for a side-by-side comparison with Revision 1.

1 The Daily IC Time-Series Framework

The Revision 1 derivations suffered from a fundamental inconsistency: Threshold 1 used an effective sample size of $N = 750$ (days) derived from the Fisher z -transformation of a pooled Spearman ρ , while Thresholds 3 and 5 used $N \approx 10^5$ (pooled stock–day pairs) in $\sigma_0^2 \approx 1/N$, and Threshold 2 oscillated between $N = 1850$ and $N = 375\,000$ depending on the argument. These inconsistent unit assumptions led to contradictory standard errors and, consequently, thresholds that did not align with the actual computation in `diagnose_v2.py`.

Revision 2 adopts a single, self-consistent unit of analysis: the *daily cross-sectional Spearman IC*.

Definition 1.1 (Daily IC time-series). For a set of N stocks and T held-out trading days, let $\hat{\rho}_t$ be the cross-sectional Spearman rank correlation between predicted returns $\{\hat{r}_{i,t}\}_{i=1}^N$ and realized returns $\{r_{i,t}\}_{i=1}^N$ on day t :

$$\hat{\rho}_t = \text{Corr}_{\text{Spearman}}(\{\hat{r}_{i,t}\}_{i=1}^N, \{r_{i,t}\}_{i=1}^N), \quad t = 1, \dots, T. \quad (1)$$

The daily IC time-series is $\{\hat{\rho}_1, \dots, \hat{\rho}_T\}$.

Empirical calibration. The pipeline v2 held-out test set ($T = 795$ days, $N = 477$ stocks) yields:

$$\mu_{\text{IC}} = \frac{1}{T} \sum_{t=1}^T \hat{\rho}_t = 0.0289, \quad (2)$$

$$\sigma_{\text{IC}} = \sqrt{\frac{1}{T-1} \sum_{t=1}^T (\hat{\rho}_t - \mu_{\text{IC}})^2} = 0.158, \quad (3)$$

$$p_+ = \frac{1}{T} \sum_{t=1}^T \mathbf{1}\{\hat{\rho}_t > 0\} = 0.570. \quad (4)$$

Note that $\sigma_{\text{IC}} = 0.158$ captures *both* the cross-sectional estimation error *and* genuine day-to-day variation in predictive quality—this is the key quantity for all subsequent derivations.

Remark 1.1 (Relationship between pooled IC and μ_{IC}). The traditional *pooled Spearman* $\text{IC}_{\text{pooled}}$ (a single Spearman ρ over all $T \times N$ stock–day pairs) and the daily-IC mean μ_{IC} are mathematically distinct statistics. In the pipeline v2 results, $\text{IC}_{\text{pooled}} = 0.0443$ while $\mu_{\text{IC}} = 0.0289$, illustrating that the pooled estimator can differ substantially from the daily-IC mean in finite samples.

Why the daily-IC framework is preferred for diagnostic thresholds:

1. The daily IC $\hat{\rho}_t$ is the quantity that actually enters the walk-forward evaluation (the code averages per-day ICs in Step 2's t -test).
2. $\text{SE}(\mu_{\text{IC}}) = \sigma_{\text{IC}}/\sqrt{T}$ is straightforward to estimate from the daily-IC time series; the analogous standard error for $\text{IC}_{\text{pooled}}$ requires Fisher z -transformation with an assumed effective sample size (the source of the Revision 1 inconsistency).
3. In the large- T limit under weak dependence, $\sqrt{T}(\mu_{\text{IC}} - \mathbb{E}[\hat{\rho}_t]) \xrightarrow{d} \mathcal{N}(0, \sigma_{\text{IC}}^2)$ by the CLT, providing rigorous inference without distributional assumptions.

In Revision 2, **all thresholds use μ_{IC} as the primary IC statistic**. The pooled $\text{IC}_{\text{pooled}}$ is retained only as a descriptive summary; all inference proceeds from the daily-IC time-series.

1.1 Standard Errors of the Daily-IC Mean

Definition 1.2 (i.i.d. standard error). If daily ICs are serially uncorrelated,

$$\text{SE}_{\text{iid}}(\mu_{\text{IC}}) = \frac{\sigma_{\text{IC}}}{\sqrt{T}}. \quad (5)$$

Definition 1.3 (Newey–West HAC standard error). To accommodate possible autocorrelation in daily ICs (e.g., due to persistent market regimes), the Newey–West [1] heteroskedasticity-and-autocorrelation-consistent (HAC) estimator is used. Let $\hat{\gamma}_k$ be the sample autocovariance at lag k :

$$\hat{\gamma}_k = \frac{1}{T} \sum_{t=k+1}^T (\hat{\rho}_t - \mu_{\text{IC}})(\hat{\rho}_{t-k} - \mu_{\text{IC}}). \quad (6)$$

With Bartlett kernel weights $w_k = 1 - k/(L + 1)$ and bandwidth L :

$$\text{SE}_{\text{NW}}(\mu_{\text{IC}}) = \frac{1}{\sqrt{T}} \sqrt{\hat{\gamma}_0 + 2 \sum_{k=1}^L w_k \hat{\gamma}_k}. \quad (7)$$

For $T = 795$, the automatic bandwidth $L = \lfloor 4(T/100)^{2/9} \rfloor = 6$ provides a reasonable default (Newey–West, 1994).

Table 1: Standard errors of μ_{IC} under alternative assumptions. Plug-in values: $\sigma_{\text{IC}} = 0.158$, $T = 795$.

Estimator	Formula	Value
i.i.d. SE	$\sigma_{\text{IC}}/\sqrt{T}$	$0.158/\sqrt{795} = 0.00560$
NW SE ($L = 6$)	Eq. (7)	0.00490–0.00650 (estimated)

For the remainder of this document, we use $\text{SE}(\mu_{\text{IC}}) = 0.00560$ (the i.i.d. value) as the baseline, noting that the Newey–West correction typically changes it by less than 15%.

2 Threshold 1: $|\text{IC}| < 0.02$ as Detection Threshold

2.1 Minimum Detectable IC

The single most important threshold in the framework is whether the observed IC exceeds the noise floor. With the daily-IC framework, this reduces to a one-sample test of $H_0 : \mu_{\text{IC}} = 0$.

Proposition 2.1 (Minimum detectable IC at $\alpha = 0.05$, 80% power). For a two-sided test,

$$\text{IC}_{\min} = (z_{\alpha/2} + z_{\beta}) \text{SE}(\mu_{\text{IC}}), \quad z_{0.025} = 1.960, \quad z_{0.20} = 0.842. \quad (8)$$

Plugging in $\text{SE}(\mu_{\text{IC}}) = 0.00560$:

$$\begin{aligned}\text{IC}_{\min} &= (1.960 + 0.842) \times 0.00560 \\ &= 2.802 \times 0.00560 = \mathbf{0.01570}.\end{aligned}\tag{9}$$

Remark 2.1 (Comparison with Revision 1). Revision 1 derived $r_{\min} \approx 0.102$ using Fisher z with $N = 750$ and 80% power, then argued *in the opposite direction* that $|\text{IC}| < 0.02$ was conservative because it was “far below” the detection threshold. The internal contradiction was: Fisher z predicts $\text{IC} = 0.02$ is undetectable ($p \approx 0.58$), yet the code’s Step 1 correctly flags $\text{IC} = 0.044$ as a PASS based on the actual SE. The daily-IC framework resolves this: the empirical $\text{SE} = 0.00560$ (not 0.0366 from Fisher z) gives $\text{IC}_{\min} = 0.0157$, which aligns naturally with the 0.02 threshold.

2.2 Power Analysis

For an observed IC of r , the t -statistic is

$$t(r) = \frac{|r|}{\text{SE}(\mu_{\text{IC}})} = \frac{|r|}{0.00560}.\tag{10}$$

Table 2: Power to detect various IC values at $\alpha = 0.05$ (two-sided). $\text{SE} = 0.00560$, $T = 795$.

IC r	$t = r / \text{SE}$	p -value (two-sided)	Power
0.005	0.89	0.37	0.14
0.010	1.79	0.07	0.45
0.016	2.80	0.005	0.80
0.020	3.57	3.6×10^{-4}	0.94
0.029	5.15	2.7×10^{-7}	>0.999
0.044	7.86	4.0×10^{-15}	>0.999

Thus the threshold $|\text{IC}| < 0.02$ identifies the regime where the signal is *barely above the detection floor*: at $\text{IC} = 0.02$, power is 94% (adequate), but at $\text{IC} = 0.01$, power drops to 45% (inadequate). The threshold provides a conservative margin over the minimum detectable IC of 0.016.

Proposition 2.2 (Threshold 1: Detection).

$$\boxed{|\text{IC}| < 0.02 \implies \text{Below detection}}\tag{11}$$

This is a principled threshold: it is $1.28 \times \text{IC}_{\min}$, providing a 28% safety margin while staying aligned with the empirical SE.

3 Threshold 2: Held-Out IC Gap

3.1 Framework

Let $\mu_{\text{IC}}^{(\text{train})}$ and $\mu_{\text{IC}}^{(\text{held})}$ be the mean daily ICs on the training and held-out sets, respectively, and

$$\Delta = \mu_{\text{IC}}^{(\text{train})} - \mu_{\text{IC}}^{(\text{held})}.\tag{12}$$

Under H_0 (no overfitting), both estimate the same population quantity, so $\mathbb{E}[\Delta] = 0$.

With the daily-IC framework, the variance of each mean is

$$\text{Var}(\mu_{\text{IC}}^{(k)}) = \frac{(\sigma_{\text{IC}}^{(k)})^2}{T_k}, \quad k \in \{\text{train}, \text{held}\}.\tag{13}$$

3.2 Bounding the Standard Error of the Gap

Since training and held-out sets are disjoint in time, their daily IC sequences are serially separated by the training/held-out boundary. Under the assumption of weak temporal dependence (autocorrelation decays within a few lags), the covariance term is negligible:

$$\text{SE}(\Delta) = \sqrt{\frac{(\sigma_{\text{IC}}^{(\text{train})})^2}{T_{\text{train}}} + \frac{(\sigma_{\text{IC}}^{(\text{held})})^2}{T_{\text{held}}} - 2 \text{Cov}(\mu_{\text{IC}}^{(\text{train})}, \mu_{\text{IC}}^{(\text{held})})}. \quad (14)$$

A *conservative* estimate uses the same σ_{IC} for both sets and assumes zero covariance:

$$\text{SE}(\Delta) \approx \sigma_{\text{IC}} \sqrt{\frac{1}{T_{\text{train}}} + \frac{1}{T_{\text{held}}}}. \quad (15)$$

With the empirical values ($\sigma_{\text{IC}} = 0.158$, $T_{\text{train}} = 1853$, $T_{\text{held}} = 795$):

$$\begin{aligned} \text{SE}_{\text{train}} &= \frac{0.158}{\sqrt{1853}} = 0.00367, \\ \text{SE}_{\text{held}} &= \frac{0.158}{\sqrt{795}} = 0.00560, \\ \text{SE}(\Delta) &= \sqrt{0.00367^2 + 0.00560^2} = \sqrt{1.347 \times 10^{-5} + 3.136 \times 10^{-5}} \\ &= \sqrt{4.483 \times 10^{-5}} = \mathbf{0.00670}. \end{aligned} \quad (16)$$

Table 3: Gap diagnostics in the daily-IC framework.

Gap Δ	$\Delta / \text{SE}(\Delta)$	One-sided p	Classification
0.005	0.75	0.23	Within noise
0.010	1.49	0.068	Marginally suspicious
0.020	2.99	0.0014	Significant overfitting
0.050	7.46	$< 10^{-13}$	Pathological

Proposition 3.1 (Threshold 2: Gap categories).

$$\boxed{\Delta < 0.005} \quad (\text{Healthy — within 1 SE}), \quad (17)$$

$$\boxed{0.005 \leq \Delta \leq 0.020} \quad (\text{Suspicious — 1–3 SE}), \quad (18)$$

$$\boxed{\Delta > 0.020} \quad (\text{Pathological — exceeds 3 SE}). \quad (19)$$

These thresholds are tighter than the Revision 1 values (0.01, 0.05) because the daily-IC SE is substantially smaller than the Fisher- z SE (which assumed $N = 750$ effective observations). The Revision 1 thresholds corresponded to 0.24 – 1.18σ ; the new thresholds are properly calibrated to 0.75 – 2.99σ .

3.3 Per-Window Gap

For the walk-forward per-window gap used in Step 5, the validation window contains $T_{\text{val}} \approx 252$ days:

$$\text{SE}(\Delta^{(w)}) \approx 0.158 \sqrt{\frac{1}{1008} + \frac{1}{252}} = 0.158 \times 0.0704 = 0.0111. \quad (20)$$

The threshold $\bar{\Delta}^{(w)} < 0.03$ (used in code) corresponds to $0.03/0.0111 = 2.70$ SE, a conservative bound.

4 Threshold 3: Coefficient of Variation of IC Across Windows

4.1 Framework

For W walk-forward windows, let the window- w IC be the mean of daily ICs within that window's validation period:

$$\text{IC}_w = \frac{1}{T_w} \sum_{t \in \mathcal{T}_w} \hat{\rho}_t, \quad w = 1, \dots, W. \quad (21)$$

The CV is

$$\text{CV} = \frac{s_{\text{IC}}}{|\bar{\text{IC}}|}, \quad \bar{\text{IC}} = \frac{1}{W} \sum_{w=1}^W \text{IC}_w, \quad s_{\text{IC}}^2 = \frac{1}{W-1} \sum_{w=1}^W (\text{IC}_w - \bar{\text{IC}})^2. \quad (22)$$

4.2 Expected CV Under $H_0 : \mu_{\text{IC}} = 0$

This is where Revision 1 contained a critical algebraic error. Using $\sigma_0^2 \approx 1/N$ with $N \approx 10^5$ (pooled pairs) produced $\sigma_0 \approx 0.003$, leading to the incorrect conclusion that $\mathbb{E}[\text{CV} \mid H_0] \gg 1$ was *obvious*. In reality, the window-IC standard error under H_0 is determined by the daily-IC framework, not by the pooled Fisher- z variance.

Proposition 4.1 (Expected CV under the null). Under H_0 , each window IC is a mean of T_w approximately independent daily ICs:

$$\text{IC}_w \sim \mathcal{N}\left(0, \frac{\sigma_{\text{IC}}^2}{T_w}\right). \quad (23)$$

For W windows of equal size $T_w = T_{\text{val}}$,

$$\bar{\text{IC}} \sim \mathcal{N}\left(0, \frac{\sigma_{\text{IC}}^2}{T_{\text{val}}W}\right), \quad s_{\text{IC}} \approx \frac{\sigma_{\text{IC}}}{\sqrt{T_{\text{val}}}}. \quad (24)$$

Since $|\bar{\text{IC}}|$ is $O_p(1/\sqrt{W})$ while s_{IC} is $O_p(1)$ (relative to $\sigma_{\text{IC}}/\sqrt{T_{\text{val}}}$),

$$\mathbb{E}[\text{CV} \mid H_0] \approx \frac{\sigma_{\text{IC}}/\sqrt{T_{\text{val}}}}{\sigma_{\text{IC}}/\sqrt{T_{\text{val}}W} \cdot \sqrt{2/\pi}} = \sqrt{\frac{\pi W}{2}} \approx 4.16 \quad (\text{for } W = 11). \quad (25)$$

Thus, under the null hypothesis of no predictive signal, we *expect* $\text{CV} \approx 4\text{--}5$, not $\text{CV} \gg 1$ as Revision 1 implied. This means:

- $\text{CV} < 1$ is already *strong evidence* against H_0 .
- $\text{CV} = 0.6$ is *conclusive* evidence of non-zero IC.
- The Revision 1 CV thresholds (< 0.3 , $0.3\text{--}0.5$, > 0.6) are actually *correct* despite the flawed derivation; the daily-IC framework provides the missing rigorous justification.

4.3 Boundary Signal Detection at $\text{CV} = 0.6$

Proposition 4.2 (CV as inverse SNR). For a genuine signal with $\mu_{\text{IC}} > 0$ and window-level variation σ_w ,

$$\text{CV} = \frac{\sigma_w}{\mu_{\text{IC}}}, \quad \text{SNR}_{\text{window}} = \frac{1}{\text{CV}}. \quad (26)$$

At $\text{CV} = 0.6$, $\text{SNR} = 1.667$.

- $\mathbb{P}(\text{IC}_w > 0) = \Phi(\text{SNR}) = \Phi(1.667) = 0.952$.
- t -statistic for $\bar{\text{IC}}$: $t = \bar{\text{IC}}/(s_{\text{IC}}/\sqrt{W}) = \sqrt{W}/\text{CV} = \sqrt{11}/0.6 = 5.53$, $p < 10^{-3}$ (one-sample).

Table 4: CV thresholds and implications.

CV	SNR	$\mathbb{P}(\text{IC}_w > 0)$	$t_{W=11}$	Classification
< 0.3	> 3.33	> 0.9996	> 11.1	Healthy
$0.3-0.5$	$2.0-3.33$	$0.977-0.9996$	$6.6-11.1$	Suspicious
$0.5-0.6$	$1.67-2.0$	$0.952-0.977$	$5.5-6.6$	Marginal pass
> 0.6	< 1.67	< 0.952	< 5.5	Failing
> 1.0	< 1.0	< 0.841	< 3.3	Noise-dominated

5 Thresholds 4 & 6: Daily IC Positive Fraction (Replaces Per-Stock Binomial)

5.1 The Problem with Per-Stock Binomial Tests

Revision 1’s per-stock binomial test assumes independent stock-level ICs. This assumption is *known to be violated* in financial data: factor exposures and sector co-movements induce positive cross-sectional correlation, which inflates the variance of N_+ and makes the binomial test *anti-conservative* (rejects H_0 too easily).

The daily-IC framework offers a cleaner, self-consistent alternative.

Remark 5.1 (Conceptual shift: stocks vs. days). The per-stock binomial test and the daily positive fraction test ask different questions. The per-stock test asks “are most stocks predictable?” — it counts how many individual equities yield a positive rank IC when averaged across all available windows. The daily test asks “are most days predictable?” — it counts the fraction of trading days on which the cross-sectional rank correlation is positive. These are complementary perspectives: a model may predict well on most days (high $\hat{p}_+$) but fail on a minority of persistently unpredictable stocks (low per-stock positive fraction). We adopt the daily framework because (a) it naturally handles the dependence structure among stocks, (b) each daily IC aggregates information from all N stocks, yielding a larger effective sample size (T vs. S), and (c) it aligns with the standard error computations used throughout this work.

5.2 Daily IC Positive Fraction

Definition 5.1 (Daily IC positive fraction).

$$p_+ = \frac{1}{T} \sum_{t=1}^T \mathbf{1}\{\hat{\rho}_t > 0\}. \quad (27)$$

This is the fraction of trading days on which the cross-sectional Spearman IC is positive.

Under H_0 (no predictive ability), daily ICs are symmetrically distributed around zero, so $p_+ = 0.5$ in expectation. Under a genuine positive signal, $p_+ > 0.5$.

5.3 Binomial Test on Daily Indicators

If daily IC signs are approximately independent,¹ then

$$T_+ = \sum_{t=1}^T \mathbf{1}\{\hat{\rho}_t > 0\} \sim \text{Binomial}(T, p). \quad (28)$$

¹Serial dependence in daily IC signs is typically much weaker than dependence in magnitudes, making the binomial approximation more defensible than for per-stock tests. The Newey–West adjustment for sign-autocorrelation is discussed below.

Under $H_0 : p = 0.5$:

$$z = \frac{\hat{p}_+ - 0.5}{\sqrt{0.25/T}} = \frac{\hat{p}_+ - 0.5}{\sqrt{0.25/795}} = \frac{\hat{p}_+ - 0.5}{0.01773}. \quad (29)$$

Table 5: Statistical significance of daily IC positive fractions ($T = 795$).

$\hat{p}_+$	T_+	z	p (one-sided)
0.500	398	0.00	0.500
0.520	413	1.13	0.130
0.550	437	2.82	0.0024
0.570	453	3.95	3.9×10^{-5}
0.600	477	5.64	8.5×10^{-9}
0.650	517	8.46	$< 10^{-16}$

5.4 Handling Autocorrelation in IC Signs

If daily IC signs exhibit serial dependence (e.g., consecutive positive days cluster during bull markets), the effective sample size for the binomial test is reduced. Using a runs-test-based correction:

$$T_{\text{eff}} = T \times \frac{2 R_+ R_-}{R_+ + R_-} \times \frac{1}{T}, \quad (30)$$

where R_+ and R_- are the numbers of runs of positive and negative signs, respectively. In the worst case of complete clustering (all positives together), $T_{\text{eff}} \approx 2$, rendering the test uninformative. In practice, financial IC sign sequences are well-mixed (runs-test $p > 0.1$ in most pipelines), so $T_{\text{eff}} \approx T$.

A more general approach uses Newey–West on the sign indicators themselves:

$$\text{SE}_{\text{NW}}(\hat{p}_+) = \frac{1}{T} \sqrt{\sum_{t=1}^T (Y_t - \hat{p}_+)^2 + 2 \sum_{k=1}^L w_k \sum_{t=k+1}^T (Y_t - \hat{p}_+)(Y_{t-k} - \hat{p}_+)}, \quad (31)$$

where $Y_t = \mathbf{1}\{\hat{p}_t > 0\}$.

5.5 Proposed Thresholds

Proposition 5.1 (Threshold 4: Daily IC positive fraction).

$$\boxed{\hat{p}_+ > 0.55} \quad (\text{Healthy} — z > 2.82, p < 0.0025), \quad (32)$$

$$\boxed{0.50 \leq \hat{p}_+ \leq 0.55} \quad (\text{Suspicious} — \text{marginal evidence}), \quad (33)$$

$$\boxed{\hat{p}_+ < 0.50} \quad (\text{Pathological} — \text{worse than chance}). \quad (34)$$

Proposition 5.2 (Threshold 6: Binary pass/fail). The code’s binary pass criterion becomes:

$$\boxed{\text{pass_daily_positive} = (\hat{p}_+ > 0.55)}. \quad (35)$$

For the pipeline v2 Ridge with $\hat{p}_+ = 0.570$, this is a comfortable PASS ($z = 3.95$, $p = 3.9 \times 10^{-5}$).

5.6 Comparison with Per-Stock Binomial

The daily-IC positive-fraction test is superior on three grounds:

1. **Consistent unit:** same daily cross-section used across all thresholds.
2. **Reduced dependence:** daily IC signs are less cross-sectionally correlated than per-stock ICs (which share market-wide factor exposures across all windows).
3. **Higher statistical power:** $T = 795$ daily observations vs. $S = 200\text{--}477$ stocks, with the added benefit that each daily IC aggregates information from all N stocks.

6 Threshold 5: $CV < 0.6$ as Pass

6.1 CV in the Daily-IC Framework

The code’s CV pass criterion is $CV(IC) < 0.6$, where IC is the vector of W walk-forward window ICs. As shown in §4, this corresponds to $SNR > 1.67$ and $\mathbb{P}(IC_w > 0) > 0.952$.

6.2 Illustrative Computation with Pipeline v2 Ridge

Pipeline v2 Ridge yields $W = 11$ window ICs with $\bar{IC} = 0.0648$, $s_{IC} = 0.0382$, and $CV = 0.590$:

$$SNR_{\text{window}} = 1/0.590 = 1.695, \quad (36)$$

$$\mathbb{P}(IC_w > 0) = \Phi(1.695) = 0.955, \quad (37)$$

$$t\text{-stat} = \sqrt{11}/0.590 = 5.62, \quad (38)$$

$$p\text{-value} = 2.2 \times 10^{-4}. \quad (39)$$

6.3 IC Signal Fraction

We define the *IC Signal Fraction* as the proportion of cross-window IC variation attributable to genuine predictive signal rather than estimation noise:

$$IC\text{-SF} = \frac{\sigma_{\text{sig}}^2}{\sigma_{\text{total}}^2}, \quad \sigma_{\text{sig}}^2 = s_{IC}^2 - \frac{\sigma_{IC}^2}{T_{\text{val}}}, \quad \sigma_{\text{total}}^2 = s_{IC}^2. \quad (40)$$

Here σ_{sig}^2 is the genuine cross-window signal variance (total window IC variance minus the estimation noise floor) and σ_{total}^2 is the observed window-IC variance.

The IC Signal Fraction is an IC-level analog of the Kelly–Malamud [3] P/T ratio, which bounds the fraction of *return*-level variance that is predictable. Unlike P/T , the IC-SF operates on rank-correlation (Spearman ρ) space and does not require specifying a trading strategy to convert ICs into return-level predictions. It should be interpreted as a diagnostic for signal stability, not as a direct estimate of predictable return variance.

Decompose the observed window-IC variance into signal and noise components:

$$\text{Var}(IC_w) = \underbrace{\text{Var}(\mu_{IC}^{(w)})}_{\text{signal}} + \underbrace{\frac{\sigma_{IC}^2}{T_{\text{val}}}}_{\text{estimation noise}}, \quad (41)$$

$$\sigma_{\text{sig}}^2 = s_{IC}^2 - \frac{\sigma_{IC}^2}{T_{\text{val}}}. \quad (42)$$

For pipeline v2 Ridge ($s_{\text{IC}} = 0.0382$, $\sigma_{\text{IC}} = 0.158$, $T_{\text{val}} = 252$):

$$\frac{\sigma_{\text{IC}}^2}{T_{\text{val}}} = \frac{0.158^2}{252} = \frac{0.02496}{252} = 9.91 \times 10^{-5}, \quad (43)$$

$$\sigma_{\text{sig}}^2 = 0.0382^2 - 9.91 \times 10^{-5} = 0.001459 - 0.000099 = 0.001360, \quad (44)$$

$$\text{IC-SF} = \frac{0.001360}{0.001459} = 0.932. \quad (45)$$

Thus over 93% of window-IC variance is attributable to genuine cross-window signal variation, not estimation noise. This aligns with the CV-based conclusion that the pipeline produces consistent, non-spurious predictions.

7 Multiple Testing Correction

The diagnostic framework applies up to 11 pass/fail checks across 5 steps. Without correction, the family-wise error rate (FWER) at $\alpha = 0.05$ is inflated.

7.1 Bonferroni Correction

Proposition 7.1 (Bonferroni-adjusted significance level). For $m = 11$ tests and a target FWER of $\alpha_{\text{FW}} = 0.05$,

$$\boxed{\alpha' = \frac{\alpha_{\text{FW}}}{m} = \frac{0.05}{11} = 0.00455}. \quad (46)$$

All individual-test p -values should be compared to α' , not 0.05 or 0.01.

Impact on thresholds:

- **Step 2 t -test:** $p < 0.00455$ (previously $p < 0.01$).
- **Step 3 binomial:** $p < 0.00455$ (previously $p < 0.01$). With $W = 11$, only 11/11 positive windows survives the correction ($P(X \geq 11) = 1/2048 \approx 0.00049$). A result of 10/11 windows positive yields $P(X \geq 10) = 12/2048 \approx 0.00586$, which exceeds both the Bonferroni threshold $\alpha' = 0.00455$ and the Holm-Bonferroni step-down threshold $\alpha'_{(3)} = 0.05/9 \approx 0.00556$.
- **Step 4 binomial:** $p < 0.00455$. For $T = 795$, this requires $\hat{p}_+ > 0.548$ (previously 0.55, essentially unchanged).
- **Step 4 t -test:** $p < 0.00455$ (previously $p < 0.01$).

The daily-IC-based Threshold 4 ($\hat{p}_+ > 0.55$) is nearly unchanged by the correction, while the per-window binomial test (requiring $p < 0.01$ on $W = 11$) becomes stricter.

7.2 Hierarchical Gatekeeping

A more powerful alternative to Bonferroni uses hierarchical testing:

1. **Steps 1–2 are gatekeepers:** the model must pass the held-out IC detection test *and* the t -test of $\mu_{\text{IC}} > 0$ at $\alpha = 0.05$.
2. **Steps 3–5 are applied only if gatekeepers pass.** Within Steps 3–5, apply Bonferroni correction for the remaining 9 tests ($\alpha' = 0.05/9 = 0.00556$).
3. **Overall verdict:** PASS requires all gatekeepers passed *and* at least 7/9 secondary tests passed (after correction).

This hierarchical structure reflects the logical dependence: if the model has no detectable signal (Steps 1–2), stability and overfitting checks are moot.

8 Implementation Checklist

8.1 Computing the Daily IC Time-Series

The daily IC array is already computed in `diagnose_v2.py` (lines 190–209) as `ho_daily_ic`. The existing code already computes `daily_ic_mean`, `daily_ic_std`, and `daily_ic_positive_frac` and stores them in the JSON output.

Extension needed: compute daily ICs for each walk-forward window’s validation period, to enable per-window daily-IC analysis.

8.2 Computing Newey–West Standard Errors

```
import numpy as np
from scipy import stats

def newey_west_se(ts, max_lag=None):
    """HAC standard error of the mean (Bartlett kernel)."""
    T = len(ts)
    if max_lag is None:
        max_lag = int(4 * (T / 100) ** (2 / 9))
    mu = np.mean(ts)
    gamma = np.zeros(max_lag + 1)
    for k in range(max_lag + 1):
        gamma[k] = np.mean(
            (ts[k:] - mu) * (ts[:T-k] - mu)
        )
    w = 1 - np.arange(1, max_lag + 1) / (max_lag + 1)
    long_run_var = gamma[0] + 2 * np.sum(w * gamma[1:])
    return np.sqrt(long_run_var / T)
```

8.3 Updating the Verdict Thresholds

The hard-coded thresholds in `diagnose_v2.py` should be replaced with data-driven equivalents computed from the daily IC time-series:

Table 6: Threshold migration from Revision 1 to Revision 3.

Check	Rev. 1 Value	Rev. 2 Value	Derivation
Step 1 pass	$ IC > 0.02$	$ IC > 0.02$	§2, Eq. (9)
Step 2 pass	$p < 0.01$	$p < 0.00455$	§7, Eq. (46)
Step 3 CV pass	$CV < 0.6$	$CV < 0.6$	§4, unchanged but re-derived
Step 3 binomial pass	$p < 0.01$	$p < 0.00455$	Bonferroni correction; requires 11/11 windows
Step 3 recency pass	$\Delta > -0.03$	$\Delta > -0.02$	§3, $\approx 2 \text{ SE}(\Delta^{(w)})$, prelim. ²
Step 4 binomial pass	$p < 0.01$	$\hat{p}_+ > 0.55$	§5, replaces per-stock
Step 4 <i>t</i> -test	$p < 0.01$	$p < 0.00455$	Bonferroni correction
Step 4 majority	$\hat{p} > 0.55$	$\hat{p}_+ > 0.55$	§5, daily framework
Step 5 gap pass	$\bar{\Delta}^{(w)} < 0.03$	$\bar{\Delta}^{(w)} < 0.02$	§3, $\approx 1.8 \text{ SE}$
Step 5 decay pass	$\Delta_{\text{half}} > -0.02$	$\Delta_{\text{half}} > -0.01$	Tighter, daily-IC calibrated; prelim. ³
Step 5 bootstrap pass	$CI_{\text{lower}} > 0.02$	$CI_{\text{lower}} > 0.016$	§2, IC_{min}

8.4 Validation Protocol

After implementing Revision 3 thresholds, verify against the pipeline v2 Ridge result ($IC = 0.065$, $CV = 0.59$, $\hat{p}_+ = 0.57$):

- The Ridge pipeline should remain a clear PASS (as expected for a well-established baseline).
- Apply to StockCTM (IC = 0.056–0.062, borderline) and MambaStock (IC ≈ 0 , expected FAIL) as sanity checks.
- Compare verdicts between Revision 1 and Revision 3 to ensure no unexpected threshold reversals for documented results.

9 Summary: Comparison with Revision 1

Table 7: Framework comparison: Revision 1 vs. Revision 3.

Aspect	Revision 1	Revision 3 (this document)
Unit of analysis	Pooled Spearman ρ over all stock-day pairs	Daily cross-sectional Spearman ρ_t time-series
SE(IC)	Fisher z : $1/\sqrt{N-3}$ with inconsistent N (750; 100K; 375K)	$\sigma_{IC}/\sqrt{T}$ with consistent $\sigma_{IC} = 0.158$, $T = 795$
Threshold derivation	1 Fisher $z \rightarrow r_{\min} = 0.102$, then <i>ad hoc</i> threshold 0.02	Power analysis $\rightarrow$ IC _{min} = 0.0157, principled threshold 0.02
Threshold 2 gap	Two opposite SE estimates (0.042 and 0.002) used in same section	Single consistent SE = 0.0067, from daily-IC σ_{IC}
Threshold 3 CV	$\sigma_0^2 \approx 10^{-5}$ (pooled pairs) $\rightarrow$ CV _{null} $\gg 1$	$\sigma_0^2 = \sigma_{IC}^2/T_w$ (daily ICs) $\rightarrow$ CV _{null} ≈ 4.2
Per-stock binomial	$S = 200$ independent stocks; ignores cross-sectional correlation	Replaced by daily-IC positive fraction $\hat{p}_+$ with $T = 795$ daily observations
Multiple testing	Not addressed	Bonferroni $\alpha' = 0.05/11 = 0.00455$; hierarchical gatekeeping alternative
Theoretical foundation	Fisher z -transformation (bivariate normality assumption)	Central Limit Theorem for daily-IC means (valid under weak dependence)

References

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